NHS Emergency Care Demand, Waiting-Time Performance and Forecasting

A reproducible analytics and business intelligence portfolio project using Python, forecasting methods and Power BI.

1. Executive summary

This project examines emergency care demand and waiting-time pressure across NHS England over seven financial years (2019/20–2025/26). Monthly A&E attendance and emergency admission data were consolidated, cleaned and analysed to describe national trends, compare NHS providers, measure four-hour waiting breaches and Decision to Admit (DTA) delays, and evaluate time-series forecasting approaches. Findings were presented in a four-page interactive Power BI report.

Across the period, the analysed monthly data contain approximately 168.96 million A&E attendances and 43.28 million emergency admissions. The attendance-weighted four-hour breach rate is approximately 22.61%, while recorded DTA waits exceeding four hours amount to 5.39 per 100 A&E attendances. These indicators describe different aspects of system pressure and should not be interpreted interchangeably.

2. Objectives and analytical questions

- Quantify national A&E demand and emergency admissions over time.
- Track four-hour waiting breaches and DTA delays by month and financial year.
- Compare NHS provider activity and waiting-time performance while applying a minimum-volume threshold.
- Evaluate forecasting models against observed monthly attendance and communicate accuracy and limitations.
- Build an interactive Power BI report that supports evidence-led interpretation.

3. Data sources and scope

The source was NHS England's monthly A&E Attendances and Emergency Admissions publication. The working project consolidated 84 monthly files covering April 2019 to March 2026. The initial consolidated data contained approximately 17,595 rows and 34 columns; the provider-focused cleaned extract contained approximately 17,511 rows and 22 columns. National monthly

aggregates and provider-by-year summaries were subsequently prepared for analysis and reporting.

The Power BI model includes Fact_AE_Monthly (84 monthly records), Fact_AE_Annual (seven financial-year records), Fact_AE_Provider (approximately 1,416 provider-year records), forecasting comparison and projection tables, model evaluation, and backtesting outputs. A Dim_Financial_Year lookup supports filtering across the national and provider visuals.

4. Data preparation and quality assurance

- Imported and combined the monthly NHS publication files into a consistent analytical structure.
- Removed aggregate total rows from the provider-level extract to avoid double-counting when analysing organisations.
- Harmonised inconsistent and duplicated four-hour waiting fields, converted numerical fields, and checked missing values in core provider records.
- Prepared month and financial-year identifiers and produced monthly, annual and provider-year reporting tables.
- Created a minimum annual attendance flag (10,000 attendances) to reduce distortion in provider comparisons involving very small volumes.
- Established Power BI relationships from the financial-year dimension to the relevant fact tables; tested slicer responsiveness and core measures.

5. Indicators and calculation logic

Indicator	Definition
Total A&E attendances	Sum of recorded attendances.
Four-hour waiting rate	Attendances exceeding four hours ÷ total A&E attendances × 100.
Total four-hour breaches	Sum of attendances recorded as exceeding four hours.
DTA waits over four hours	Sum of recorded waits exceeding four hours following a decision to admit.
DTA waits per 100 attendances	DTA waits over four hours ÷ total A&E attendances × 100.
Total emergency admissions	Sum of emergency admissions in the national monthly reporting table.

NHS provider count	Distinct count of provider organisation codes.

Interpretation note: the DTA-per-100-attendances indicator is a normalised count, not the percentage of admitted patients delayed. Four-hour rates and DTA ratios are already scaled to 0–100 in the project measures; applying an additional percentage format would misstate them.

6. National findings

Indicator	Reported result
A&E attendances, 2019/20–2025/26	168.96 million
Emergency admissions, 2019/20–2025/26	43.28 million
Four-hour waiting rate, whole period	22.61%
DTA waits >4h per 100 attendances, whole period	5.39
A&E attendances, 2019/20	25.02 million
A&E attendances, 2020/21	17.13 million
A&E attendances, 2025/26	26.97 million
Four-hour waiting rate, 2025/26	25.63%
DTA waits >4h per 100 attendances, 2025/26	5.83

The national attendance series shows a marked fall in 2020/21, followed by recovery and an increase to approximately 26.97 million attendances in 2025/26. The four-hour breach rate increased substantially after 2020/21, reaching 27.93% in 2023/24 before easing to 25.63% in 2025/26. The later improvement should not obscure the continued scale of waiting-time pressure.

7. Provider-level analysis

Provider performance was assessed through distinct organisation counts, attendance volumes, four-hour breach rates and DTA waits per 100 attendances. Top-10 horizontal bar charts rank providers by waiting rate and attendance volume; an accompanying table brings the indicators together. The table uses conditional colour bands (below 15, 15–under 30, and 30 or above) to aid visual scanning. These bands are project-specific display categories, not official NHS standards.

Provider rankings are most interpretable when a single financial year is selected. Multi-year aggregation may mix different provider-year circumstances; comparisons are also sensitive to differences in provider type and activity mix. The 10,000-attendance threshold is applied to the comparison visuals to support more meaningful interpretation.

8. Forecasting methodology and evaluation

Monthly attendance forecasting was explored using Holt-Winters, Seasonal Naive, Simplified SARIMA and Original SARIMA approaches. Predictions were compared with observed attendances over April 2025–March 2026. Evaluation used Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), with additional backtesting outputs prepared in the analytical workflow.

Forecasting model	MAPE (2025/26)
Holt-Winters	1.564%
Simplified SARIMA	2.403%
Seasonal Naive	2.405%
Original SARIMA	30.516%

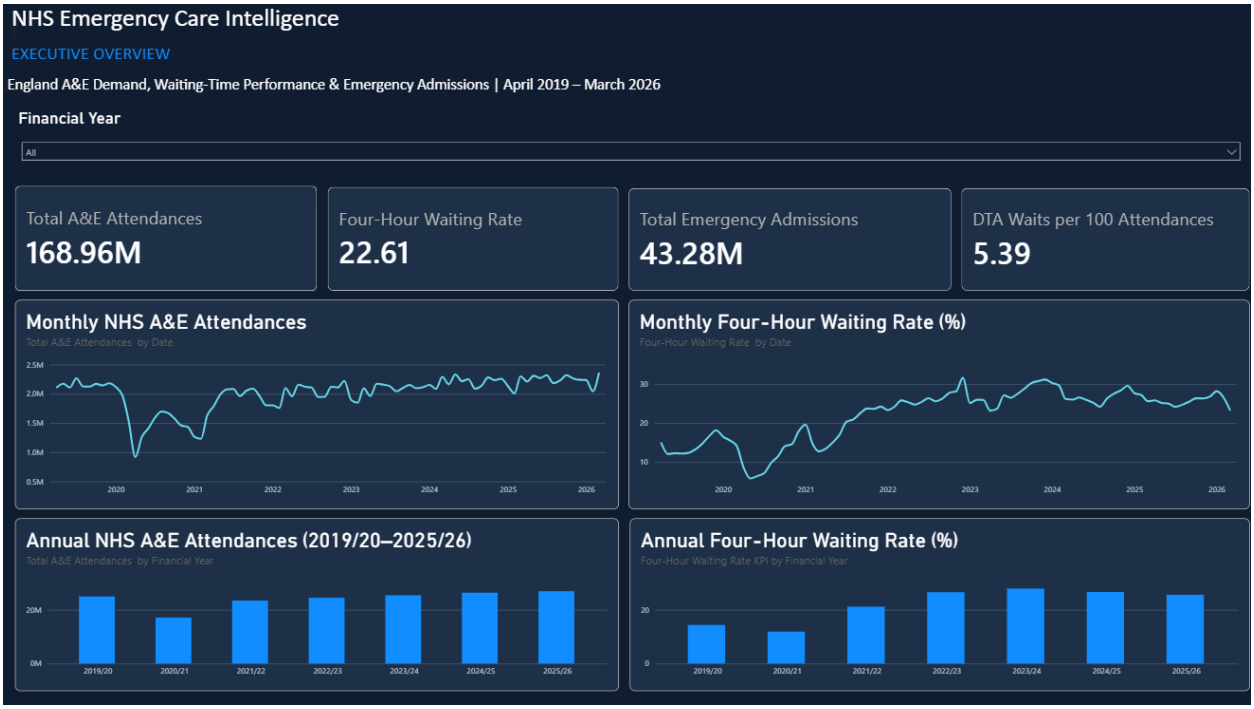
Holt-Winters had the lowest MAPE in the reported comparison, indicating the closest average percentage agreement with actual monthly attendance among the tested models. The original SARIMA configuration performed substantially worse in this comparison. RMSE was also visualised to make larger attendance-volume errors more visible.

Critical limitation: the 2025/26 comparison period was also used during model tuning. The figures must not be presented as an independent, untouched holdout evaluation. Although additional backtesting was undertaken, model selection and generalisation claims require careful distinction between tuning and evaluation periods. The forecast tables currently displayed in Power BI cover April 2025–March 2026; they are historical-period predictions, not projections beyond March 2026.

9. Power BI deliverables

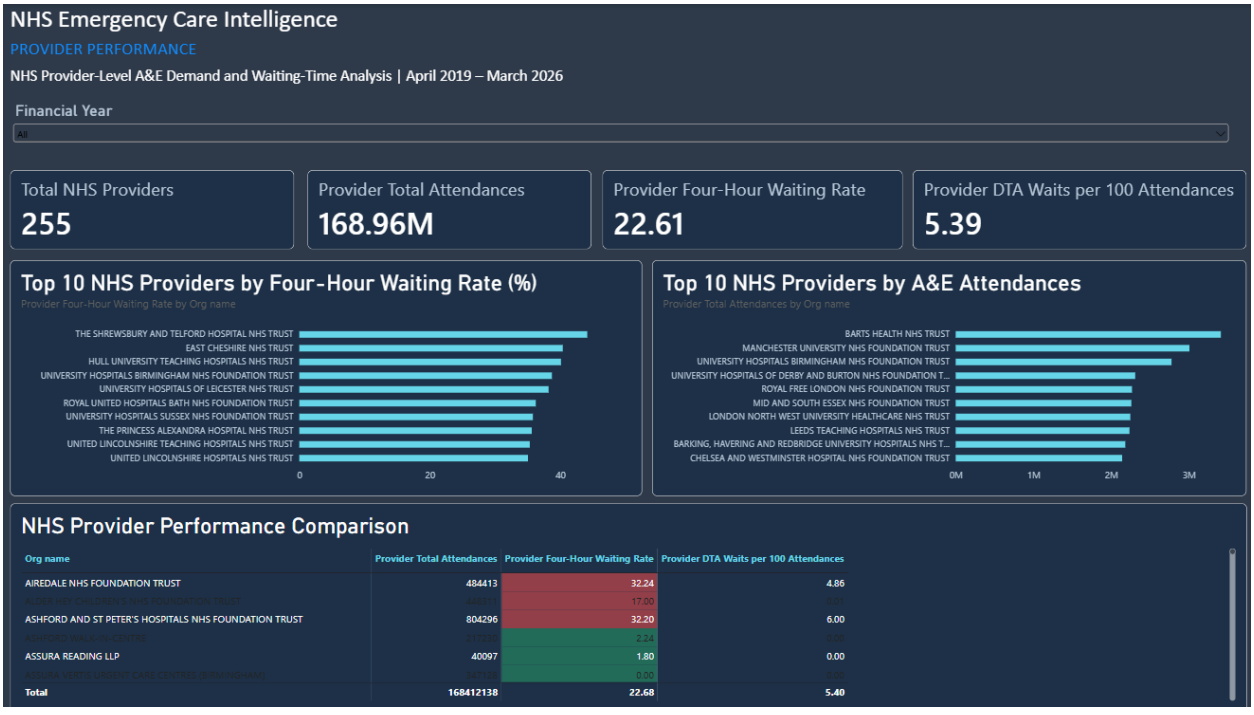
Executive Overview

Four national KPIs, a financial-year slicer, monthly attendance and waiting-rate trends, and annual attendance and waiting-rate comparisons.



Provider Performance

Four provider KPIs, a financial-year slicer, top-10 provider comparisons by breach rate and attendance, and a colour-coded performance table.



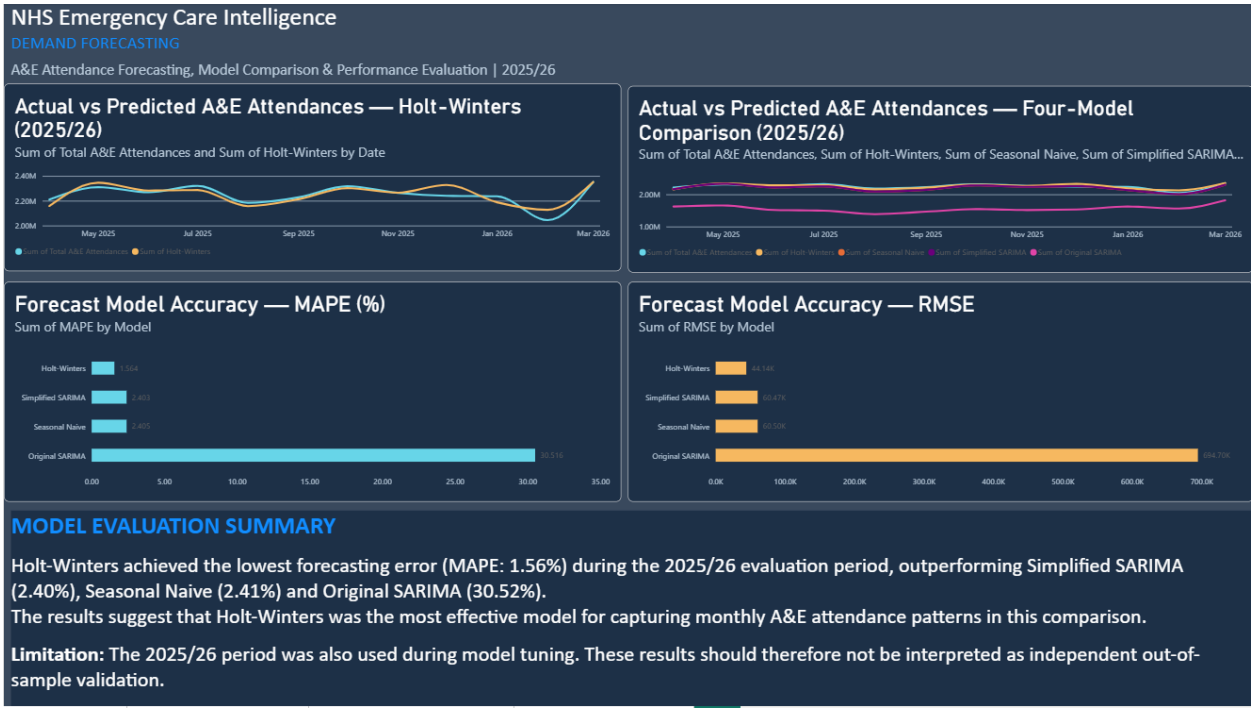
Emergency Care Pressure

Four pressure KPIs, a financial-year slicer, monthly four-hour breach and DTA trends, and annual comparisons of these indicators.



Demand Forecasting

Holt-Winters actual-versus-predicted chart, four-model comparison, MAPE and RMSE ranking charts, and a written model evaluation summary.



The report uses a consistent dark-navy theme with teal and amber accents. The working Power BI file is NHS_Emergency_Care_Intelligence.pbix.

10. Tools and technical skills demonstrated

Python and pandas for file consolidation, data cleaning, aggregation and time-series preparation; statistical forecasting methods for attendance prediction and error evaluation; Power BI, data modelling and DAX for interactive measures, filtering, visual comparison and presentation; and quality-assurance practices for reconciling reporting levels and clarifying metric definitions.

11. Limitations and recommended next steps

- Retain the distinction between attendance-based waiting rates and DTA wait counts per 100 attendances.
- Interpret provider comparisons in context, including provider type, size and financial-year selection.
- Document training, tuning and backtesting windows explicitly; evaluate selected models on genuinely untouched later data where available.
- Generate and label post-March-2026 forecasts separately if a genuine future-demand projection is required.

- Review NHS England data reuse terms and avoid committing unnecessarily large raw files or sensitive information.

12. Conclusion

The project integrates seven years of NHS England emergency care data into an analytical workflow spanning data engineering, descriptive statistics, provider benchmarking, forecasting and business intelligence. The final four-page Power BI report makes national demand, waiting pressures and model performance accessible in one place. The strongest reported forecasting performance came from Holt-Winters, while the evaluation design and the absence of post-March-2026 projections remain important qualifications. The project provides a practical demonstration of data preparation, analytical judgement, DAX modelling and communication of evidence for health-service decision support.

Source and verification note

Primary data source: NHS England, A&E Attendances and Emergency Admissions, monthly statistical publications (April 2019–March 2026). Numerical values and implementation details in this report reflect the project results reviewed during development; they have not been independently recalculated from the original CSV files for this document. Verify final published figures against the saved analysis outputs before external publication.

Report written by:

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